

Adaptive Human-AI Authority Allocation

Threshold learning, personalization, calibration, and longitudinal collaboration

Tier 2 - Master's Research Design

Human-AI Codecision Systems Research Program

Stage One revision - September 2026

Evidence status: Synthetic Stage One contains five independent seeds, 180 trials per policy and seed, 3,600 total decisions, and a controlled AI distribution shift in the final quarter. There were zero human participants. Adaptive codecision selectively regulated authority but did not improve synthetic accuracy or declared utility over human-only control. Human trust, satisfaction, and real cognitive state remain unmeasured.

Abstract

Stage One demonstrates selective intervention but not improved outcomes. The graduate extension therefore asks whether authority allocation can be learned or personalized rather than fixed by one threshold configuration. The study varies simulated human expertise, fatigue, AI skill, distribution shift, uncertainty thresholds, confidence thresholds, and critical-risk thresholds before any human-participant study is attempted. Only after a computational policy is stable and auditable should a separately approved human study evaluate trust, workload, reliance, and satisfaction.

1. Research program

Factor	Planned treatment
Human expertise / fatigue	controlled synthetic sweeps
AI skill	low-to-high calibrated sweep
Uncertainty threshold	multi-level ablation
AI-confidence threshold	multi-level ablation
Critical-risk threshold	multi-level ablation
Distribution shift	timing and severity blocks
Policy adaptation	fixed threshold vs learned/personalized policy

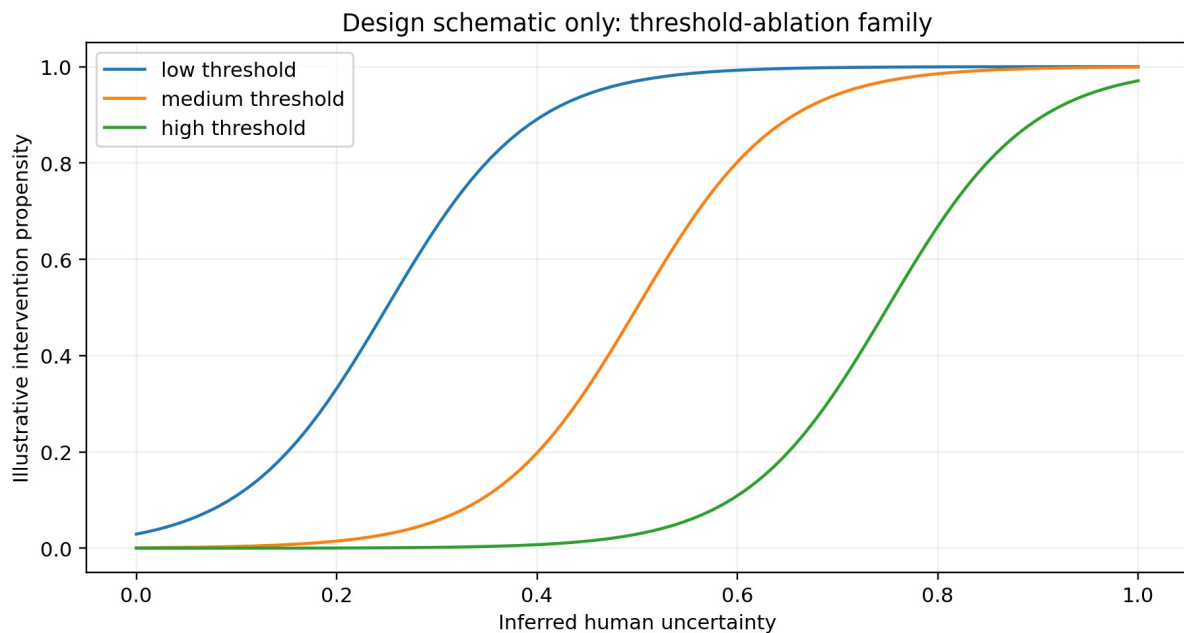


Figure 1. Design schematic only. Curves illustrate threshold families; they are not measured Stage One outcomes.

2. Primary outcomes

- Decision accuracy and declared collaborative utility.
- Escalation, override, and unnecessary-override rates.
- Autonomy/authority allocation and contestability use.

- Calibration under AI distribution shift.
- Robustness across simulated human and AI skill combinations.
- For a later approved participant study only: trust calibration, reliance, workload, satisfaction, and perceived autonomy.

3. Separation of computational and human evidence

Computational experiments should first identify stable candidate policies without using human-participant claims. A subsequent participant protocol must independently validate whether behavioral telemetry predicts uncertainty and whether intervention timing changes trust or workload. Synthetic-agent results must never be pooled with participant records as though they were one population.

4. Graduate contribution

The target contribution is a reproducible map of when adaptive authority allocation is useful, neutral, or harmful across agent-skill and threshold regimes, followed - only if ethically approved - by human validation of selected policies. A finding that human-only or static recommendation remains superior in many regimes is scientifically admissible.